END SEMESTER EXAMINATION · OCTOBER 2026

Machine Learning

B.Tech. Computer Science and Engineering · Semester V

DATE	DURATION	MAXIMUM MARKS	QUESTIONS
28 October 2026	3 Hours	100	20

Name _____ Roll No. _____

INSTRUCTIONS

1. All questions are compulsory.
2. Marks are shown against each question.
3. Section A: ten MCQs, 2 marks each.
4. Sections B and C: five 4-mark and five 12-mark questions.
5. Answer all parts together in the answer booklet.
6. Non-programmable calculator permitted; show all steps.

Section A — Multiple Choice Questions — 20 marks

Attempt all questions. Each question carries 2 marks.

1. Which statement best describes the bias–variance trade-off? [2]

(a) Increasing capacity always reduces both bias and variance.
(b) High bias causes underfitting; high variance causes overfitting.
(c) Regularisation raises variance and lowers bias.
(d) Adding training data raises variance without affecting bias.

2. Under the naive Bayes assumption, the features are: [2]

(a) independent given the class
(b) independent of the class
(c) dependent on the evidence
(d) always normally distributed

3. In the Bayesian network A → B ← C, observing B makes A and C: [2]

(a) independent
(b) d-separated
(c) conditionally dependent
(d) marginally dependent

4. In alpha–beta pruning, a branch can be pruned when its value is: [2]

(a) worse than an ancestor's best available option
(b) equal to the value of its parent node
(c) better than every sibling subtree
(d) undefined at a MAX node

5. Logistic regression with weights w and bias b predicts probabilities using: [2]

(a) $\sigma(w \cdot x + b)$
(b) $w \cdot x + b$
(c) $\exp(w \cdot x)$
(d) $\max(0, w \cdot x + b)$

6. For a split of a node into children, information gain is: [2]

(a) H(parent) − H(children)
(b) H(children) − H(parent)
(c) H(parent) + H(children)
(d) H(parent) / H(children)

7. The k-means algorithm seeks to minimise: [2]

(a) the maximum distance between any two points
(b) the sum of squared distances to assigned centroids
(c) the number of non-empty clusters
(d) the between-cluster variance

8. In a soft-margin SVM, increasing the penalty parameter C generally: [2]

(a) widens the margin and permits more violations
(b) narrows the margin and penalises violations more
(c) removes the need for slack variables
(d) forces the kernel to become linear

9. Principal component analysis projects data onto directions that: [2]

(a) maximise class separation
(b) maximise retained variance
(c) minimise the feature count
(d) maximise the number of clusters

10. Repeated k-fold cross-validation is used mainly to: [2]

(a) remove all overfitting from a model
(b) estimate generalisation performance more reliably
(c) increase the size of the training set
(d) choose the number of folds automatically

Section B — Short Answer Questions — 20 marks

Attempt all questions. Each question carries 4 marks.

11. State Bayes' theorem for events A and B. Hence express $P(B|\neg A)$ in terms of the prior and the likelihoods, and explain why the denominator acts as a normaliser. [4]

12. Explain d-separation in a Bayesian network. For the graph $X \to Z \leftarrow Y$, state whether X and Y are d-separated when Z is observed. Justify your answer. [4]

13. Compare L_1 and L_2 regularisation. Write the penalties $\|w\|_1$ and $\|w\|_2^2$, and explain one practical difference between the resulting parameter estimates. [4]

14. Define precision and recall for a binary classifier at decision threshold $\tau \in [0, 1]$. Explain the trade-off between them as τ is varied. [4]

15. For $f(x) = e^{-x} + \sqrt{x}$ on the domain $[0, \infty)$, write one gradient-descent update for x. State one condition on the learning rate η for stable convergence. [4]

Section C — Long Answer Questions — 60 marks

Attempt all questions. Each question carries 12 marks. Answer all parts together.

16. Consider a multinomial naive Bayes spam filter with vocabulary size V and class token totals N_{spam} and N_{ham}. [12]

(a) Derive the decision rule from Bayes' theorem and state the conditional independence assumption. Write the add-one smoothed estimate of $P(w|y)$. [6]

(b) For a document with $n_{free}=3$, $n_{offer}=1$, $V=2000$, $N_{spam}=N_{ham}=1000$ and equal priors, compute the likelihood ratio $P(\text{document}|\text{spam})/P(\text{document}|\text{ham})$. [6]

17. Consider the Bayesian network with edges $A \to C$, $B \to C$, $C \to D$, and $C \to E$. [12]

(a) Define d-separation and the Markov blanket. Give the Markov blanket of C. Are A and B independent given C? Given C and D? Justify. [6]

(b) Explain why exact inference is hard in general Bayesian networks and describe one approximate inference method. [6]

18. Consider a binary dataset S with 10 examples: 6 positive and 4 negative. [12]

(a) Define entropy and information gain. Compute H(S) and the gain for a split into S_1=(4+,1−) and S_2=(2+,3−). Leave logarithms exact. [6]

(b) State and explain two techniques for reducing overfitting in decision trees. [6]

19. Consider a two-layer network $h = \sigma(W_1 x + b_1)$, $o = W_2 h + b_2$, with softmax output p and cross-entropy loss L. [12]

(a) Write the forward pass and loss. Derive $\partial L/\partial W_1$, identifying where $\sigma'(z)$ appears. [6]

(b) Explain the vanishing-gradient problem in deep sigmoid networks and state one remedy. [6]

20. Consider a dataset with 1,000 samples and 50 numerical features. [12]

(a) Compare k-means and a GMM in assumptions, assignment, and objective. State how PCA selects components and the variance retained by k components. [6]

(b) Describe nested cross-validation for comparing three tuned models. Why is it less optimistic than reusing the same folds? [6]

— End of question paper —